

1. Introduction

Airline punctuality and reliability are important indicators of operational performance, while flight delays are associated with passengers' perceptions of service quality and satisfaction (Oliveira et al., 2023). Evaluating these outcomes is inherently multidimensional, as performance varies across time, airports, routes, airlines, flight volumes, on-time departures and arrivals, and cancellation levels. Examining these factors separately may obscure relationships and irregularities that become more visible when multiple dimensions are explored together.

The Bureau of Infrastructure and Transport Research Economics (BITRE) publishes punctuality and reliability data for major domestic and regional airlines operating between Australian airports (Bureau of Infrastructure and Transport Research Economics [BITRE], n.d.). The dataset used in this project covers monthly domestic aviation performance from January 2010 to February 2024. Interactive multidimensional visual analytics is applied to support both high-level monitoring and detailed exploration of geographic, temporal, route, and airline performance patterns.

The project aims to examine the geographic distribution of airport punctuality and flight activity, analyse changes in airport and airline performance across years, identify monthly patterns and potential seasonal variation, and compare the punctuality and reliability of routes and airlines. Interactive visualisation is further used to identify notable irregularities and explore how different operational dimensions collectively shape observed performance patterns.

2. Data and Analytical Preparation

2.1 Dataset Overview

The Airline On-Time Performance dataset was obtained from BITRE's domestic on-time performance reporting program, which publishes monthly punctuality and reliability statistics for Australian domestic aviation (BITRE, n.d.). A preliminary data audit was conducted in R using the `readxl` and `dplyr` packages to inspect worksheet structure, column names, date coverage, record counts, and the number of unique airports, routes, and airline labels. The audit identified 12 worksheets and confirmed that the standalone 2020 worksheet duplicated the 2020 records already included in the 2020–23 worksheet. Following integration and cleaning, the final dataset contained 80,972 monthly observations covering January 2010 to February 2024, representing 170 monthly periods, 42 physical airports, 155 route labels, and 14 raw airline labels.

The principal dimensions were month, route, departing port, arriving port, and airline. Operational measures included sectors scheduled and flown, cancellations, on-time and delayed departures and arrivals, and the corresponding punctuality and cancellation percentages. These dimensions allow geographic, temporal, and operational performance to be analysed jointly.

2.2 Data Cleaning and Integration

Data cleaning and integration were performed in R using the `readxl` and `dplyr` packages. The initial workbook contained 12 worksheets. A duplicate-key audit confirmed that all 4,197 observations in the standalone 2020 worksheet were already represented in the 2020 records of the 2020–23 worksheet; therefore, the

standalone sheet was excluded before integration to prevent double counting. The remaining 11 worksheets were combined, producing 80,977 rows, from which five non-data rows with invalid route or month fields were removed.

Raw column headers were standardised, text fields were trimmed, the monthly field was converted to a consistent date type, and operational measures were converted to numeric variables. Airline naming inconsistencies were also standardised, reducing 14 raw labels to 12 consistent analytical labels. The cleaning process identified 91 missing departure and arrival on-time performance values. Validation confirmed that all occurred where zero sectors were flown, so these undefined percentages were retained as missing rather than replaced with zero. No duplicated analytical keys or material flight-count inconsistencies remained after validation.

2.3 Analytical Measures and Data Layers

To support analysis at different levels of detail, an `Analysis_Level` variable was created in R. The 80,972 observations were classified into four analytical layers: 170 network-month records, 1,174 airline-month records, 20,156 route-month records, and 59,472 airline-route-month records. The network layer provides an overall monthly view of the aviation network. The airline layer supports comparisons between airlines, the route layer is used for route and airport analysis, and the airline-route layer provides a finer level of detail for potential airline–route analysis. The published punctuality and cancellation percentages were also checked by recalculating them from the underlying flight counts. The recalculated results were consistent with the original percentages, with only negligible numerical differences caused by computer rounding. For aggregated analysis, departure OTP was

calculated by dividing total on-time departures by total sectors flown and multiplying by 100. Arrival OTP was calculated in the same way using total on-time arrivals. Cancellation rate was calculated by dividing total cancellations by total sectors scheduled and multiplying by 100.

These weighted calculations were used instead of simple averages of percentage values because flight volumes differ between observations. This prevents low-volume records from having the same influence as high-volume operations.

3. Visualisation Design and Technical Approach

3.1 Visual Analytics Design Strategy

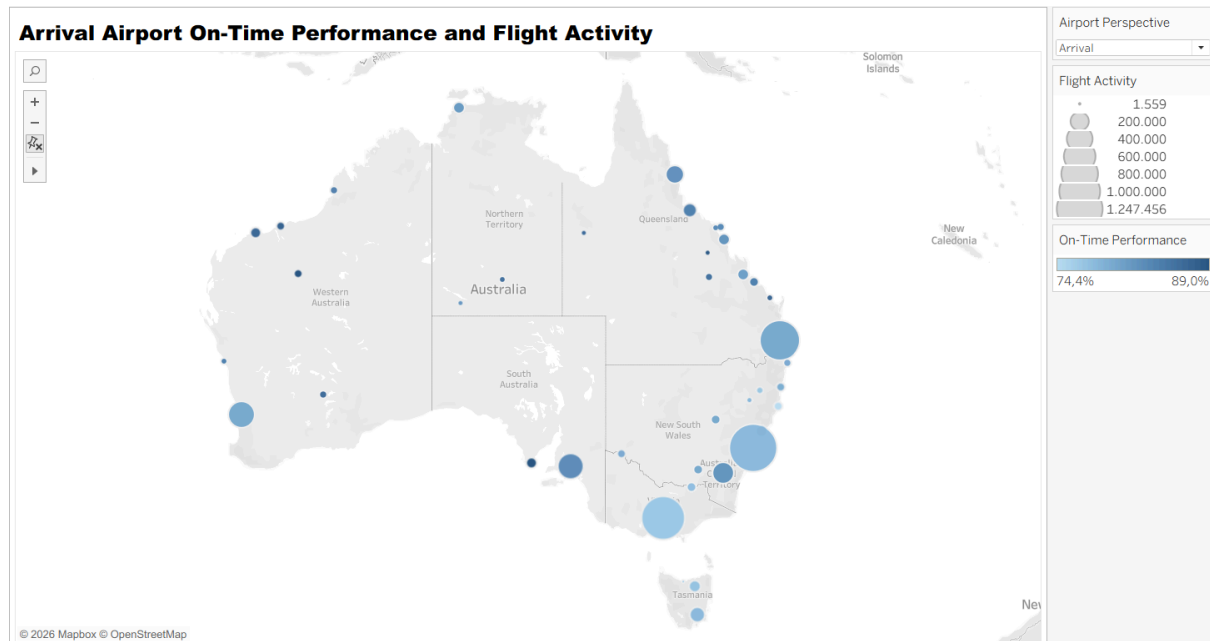
The visualisation design followed an overview-to-detail structure, progressing from network-wide performance to airport geography, temporal variation, routes, and airlines. This sequence is conceptually consistent with Shneiderman's (1996) overview-first and details-on-demand approach to interactive information visualisation. Rather than combining all dimensions in a single display, seven Tableau worksheets were developed, with each view addressing a specific analytical task while using consistent weighted performance measures. The Airport Perspective parameter and interactive tooltips support exploration, while worksheet-level year filters define the analytical scope of specific views. This structure reduces visual clutter while preserving multidimensional analysis across geography, time, flight volume, punctuality, and cancellations.

3.2 Geographic Airport Visualisation

An interactive proportional-symbol map was developed to examine airport performance while preserving geographic context. Airport location was represented by geographic position, bubble size encoded total sectors flown, and colour represented weighted on-time performance. An Airport Perspective parameter allowed the same visualisation to switch dynamically between departure and arrival performance by changing both the airport field and the corresponding weighted OTP measure. Tableau successfully geocoded most airport locations after Australia was specified as the country context; Mildura, Ayers Rock, and Hamilton Island required manual coordinate matching. Interactive tooltips provide exact airport, flight activity, and punctuality values. This design enables operational scale and performance to be explored simultaneously across geographically dispersed airports.

Figure 1

Geographic Distribution of Australian Airport On-Time Performance and Flight Activity



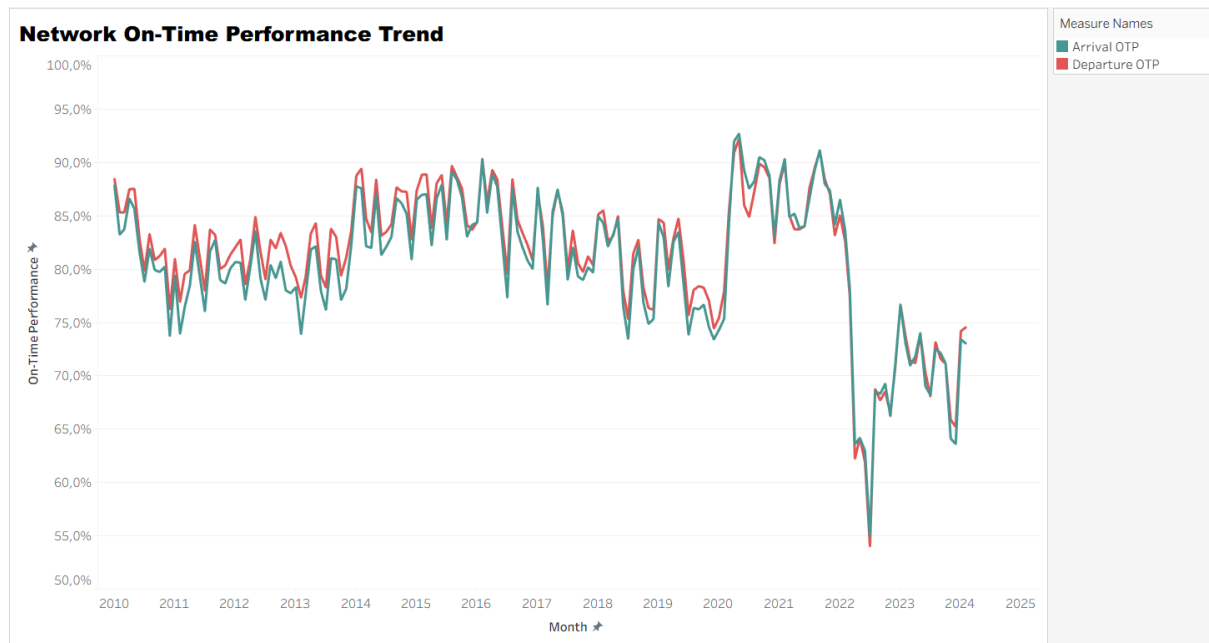
Note. Bubble size represents total sectors flown, while colour represents weighted on-time performance. The Airport Perspective parameter switches the visualisation between departure and arrival airport performance.

3.3 Temporal and Monthly Performance Visualisation

Long-term network performance was examined using a line chart that plots weighted departure and arrival OTP from January 2010 to February 2024. Displaying both measures on the same temporal axis preserves monthly continuity and enables direct comparison between departure and arrival punctuality over time, as shown in Figure 2.

Figure 2

Network On-Time Performance Trend From January 2010 to February 2024



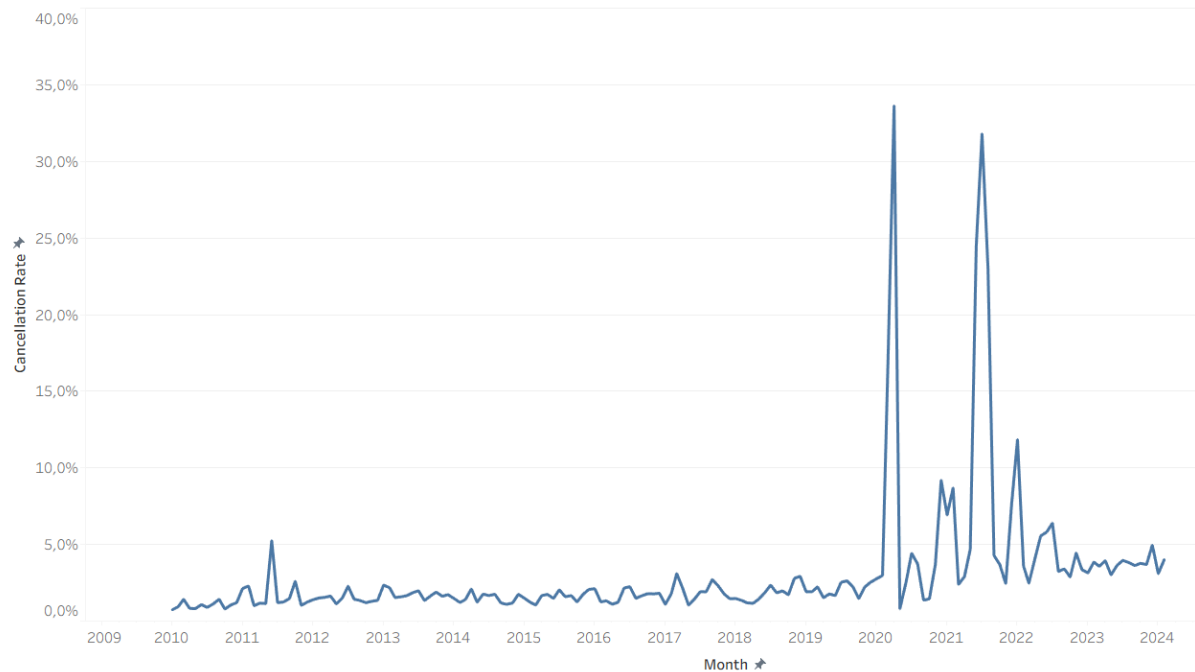
Note. The lines represent weighted departure and arrival on-time performance calculated from sectors flown. Monthly observations are shown from January 2010 to February 2024.

Cancellation rate was displayed in a separate time-series view because its different scale and sharp irregularities would reduce the readability of the punctuality measures if combined within the same chart. Figure 3 therefore provides a focused view of changes in network cancellation reliability.

Figure 3

Network Cancellation Rate Trend From January 2010 to February 2024

Network Cancellation Rate Trend

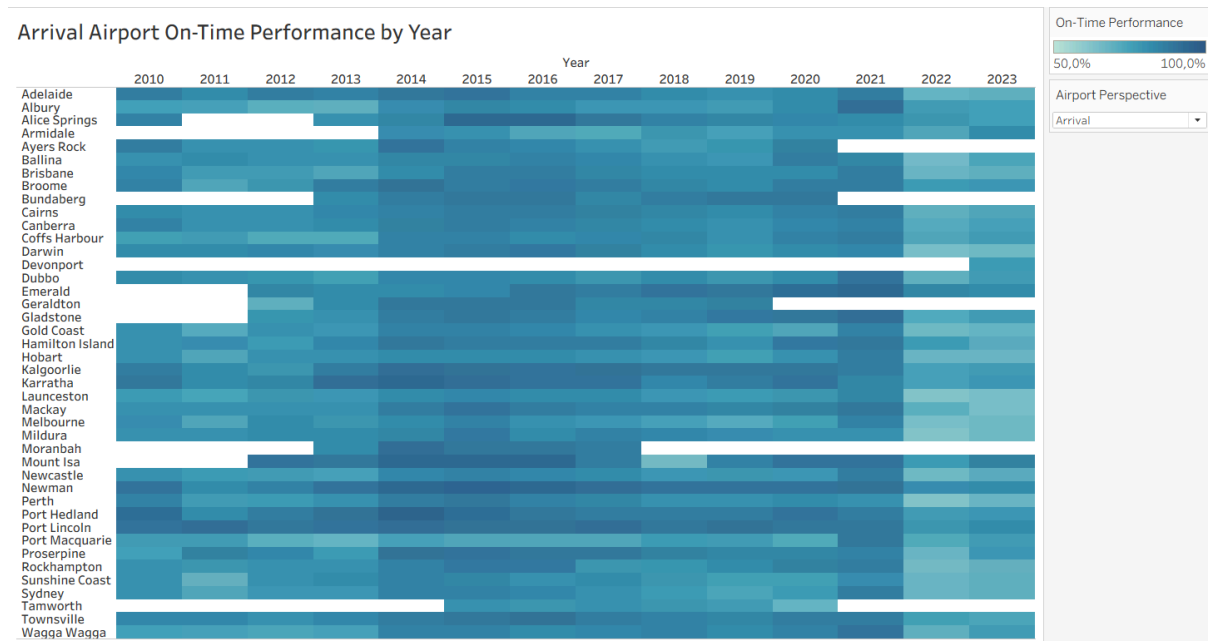


Note. Cancellation rate is calculated as total cancellations divided by total sectors scheduled. A separate scale preserves the visibility of temporal irregularities in network cancellations.

To compare many airports across time, an airport-by-year heatmap was developed using colour intensity to encode weighted OTP. The annual view covers the complete years from 2010 to 2023; partial 2024 data were excluded because the dataset contains only January and February for that year. A fixed 50%–100% colour scale supports consistent comparison across airport–year combinations (Figure 4).

Figure 4

Arrival Airport On-Time Performance by Year

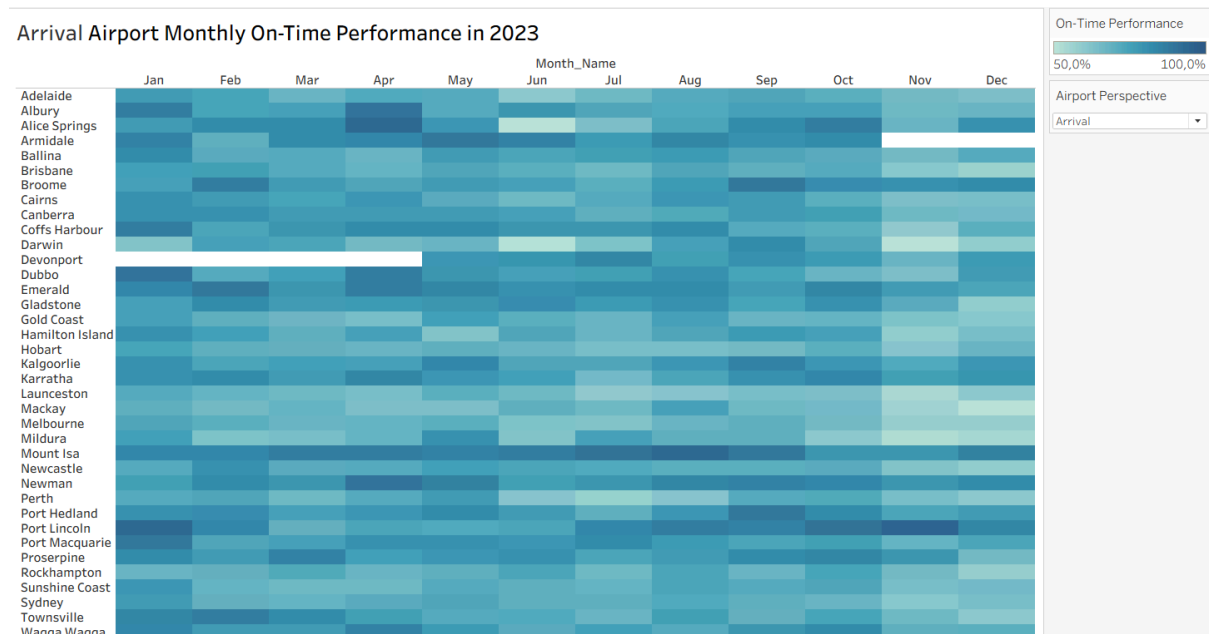


Note. Colour intensity represents weighted airport on-time performance using a fixed 50%–100% scale. White cells indicate no observation for the corresponding airport–year combination. The Airport Perspective parameter allows departure and arrival performance to be explored interactively.

Finally, Figure 5 provides a month-level heatmap for January to December 2023, the most recent complete calendar year in the dataset. Months are ordered chronologically, enabling monthly variation and potential seasonal structure to be visually explored across a complete calendar year. The same colour scale and Airport Perspective parameter were retained to maintain visual and analytical consistency with the annual heatmap.

Figure 5

Arrival Airport Monthly On-Time Performance in 2023



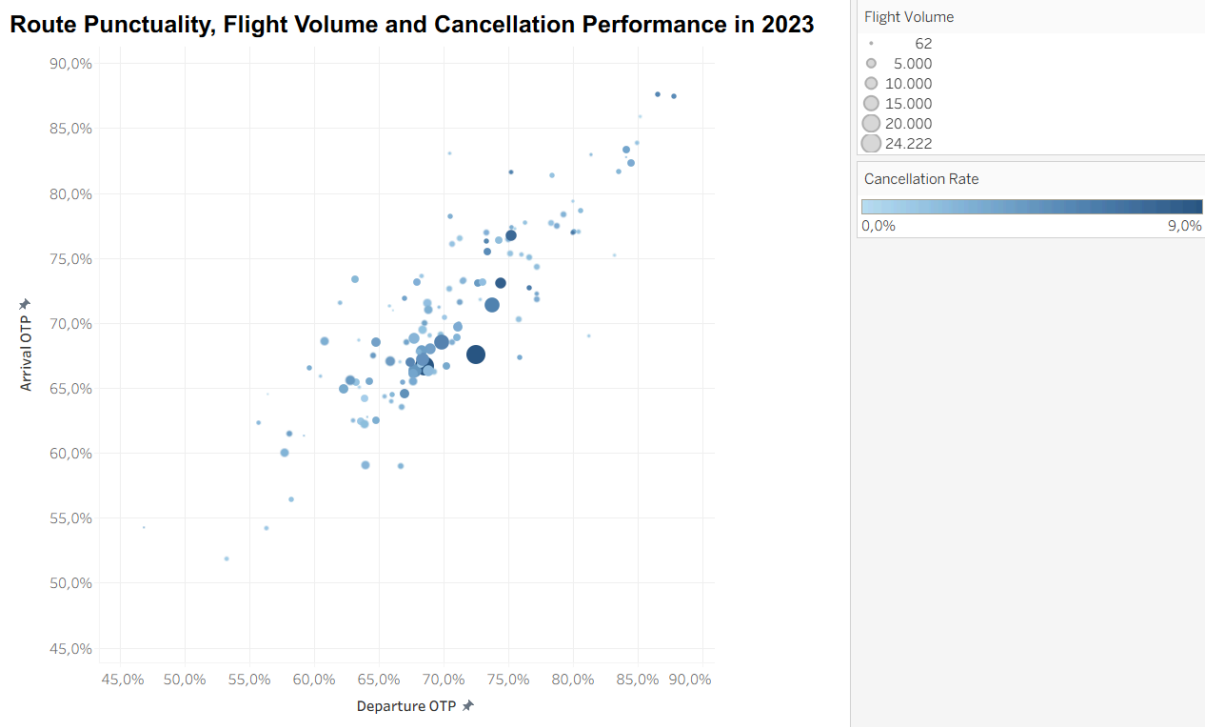
Note. Columns represent calendar months from January to December 2023, and colour intensity represents weighted airport on-time performance. White cells indicate no observation for the corresponding airport–month combination. The Airport Perspective parameter switches between departure and arrival performance.

3.4 Multidimensional and Interactive Visualisation

Route performance in 2023 was examined using a multidimensional scatter plot. Departure OTP was positioned on the horizontal axis and arrival OTP on the vertical axis, while bubble size represented sectors flown and colour encoded weighted cancellation rate. Each mark represented an individual route. Consequently, a single visual mark simultaneously communicated route identity, departure punctuality, arrival punctuality, operational scale, and cancellation reliability. Interactive tooltips provided the exact values underlying each mark, supporting closer inspection of unusual routes and dense areas of the distribution (Figure 6).

Figure 6

Route Punctuality, Flight Volume, and Cancellation Performance in 2023

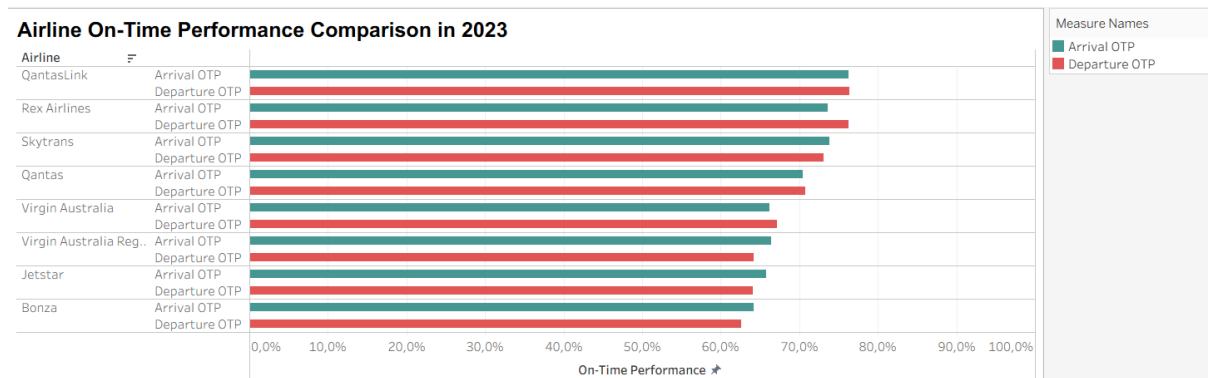


Note. Each mark represents a route. Horizontal and vertical positions indicate weighted departure and arrival OTP, respectively. Bubble size represents sectors flown, while colour represents weighted cancellation rate.

Airline performance was displayed separately using grouped horizontal bars because the relatively small number of airline categories produced a sparse scatter plot with limited comparative value. Figure 7 places departure and arrival OTP side by side for each airline and orders airlines by departure punctuality. This design supports more precise category comparison, while tooltips retain cancellation rate and flight volume as additional operational measures. The route scatter plot therefore emphasises multidimensional relationships and irregularities, whereas the airline bar chart prioritises direct ranking and comparison.

Figure 7

Airline On-Time Performance Comparison in 2023



Note. Grouped bars compare weighted departure and arrival OTP for each airline. Airlines are ordered by departure OTP. Interactive tooltips provide cancellation rate and sectors flown as additional performance context.

4. Analysis and Findings

All numerical findings in this section were derived in R from the final analytical dataset using the analytical layers and weighted aggregation methods defined in Section 2.3.

4.1 Overall Network Performance

Across January 2010 to February 2024, the network recorded 7.27 million scheduled sectors and 7.08 million sectors flown, with weighted departure and arrival OTP of 81.0% and 79.9%, respectively, and an overall cancellation rate of 2.62%. Figure 2 shows substantial temporal instability after 2020. Scheduled sectors fell from 566,889 in 2019 to 228,775 in 2020, while annual cancellation rates reached 7.2% in 2020 and 10.6% in 2021. Monthly cancellation peaks occurred in April 2020

at 33.6% and July 2021 at 31.8% (Figure 3). Punctuality subsequently deteriorated sharply, reaching 54.0% for departures and 55.0% for arrivals in July 2022.

Annual OTP improved in 2023, and the first two months of 2024 showed further improvement, although performance remained below the higher levels observed before the 2022 decline. The timing of the earlier disruption overlaps with COVID-19-related restrictions; BITRE reports that publication of monthly domestic on-time performance statistics was suspended from April to December 2020 because travel restrictions affected activity levels (BITRE, n.d.). However, the analytical dataset itself does not establish the specific operational causes of the observed performance changes.

4.2 Geographic Performance of Departing and Arriving Airports

Geographic flight activity was strongly concentrated around the largest airports shown in Figure 1. Sydney, Melbourne, and Brisbane recorded 1.25 million, 1.02 million, and 863,020 departing sectors, respectively, together accounting for 56.6% of departure flight activity. However, high operational volume did not correspond directly to the strongest punctuality. Departure OTP was 79.1% at Sydney and 79.5% at Melbourne, compared with 84.1% at Adelaide and 83.4% at Perth. Port Lincoln recorded the highest departure OTP at 89.4%, while Newman achieved the highest arrival OTP at 89.0%, although both operated at substantially lower volumes.

Geographic performance also differed between departure and arrival perspectives. Sunshine Coast recorded the lowest departure OTP at 69.3% but an arrival OTP of 77.5%, whereas Port Macquarie had the lowest arrival OTP at 74.4%. These differences demonstrate that airport performance should be interpreted jointly

through flight volume, geographic context, and both departure and arrival punctuality rather than through a single OTP ranking.

4.3 Airport Performance Across Years

Figure 4 indicates that the deterioration in 2022 was broadly distributed rather than limited to a small number of airports. Network arrival OTP fell from 86.2% in 2021 to 68.8% in 2022, and all 36 airports observed in both years recorded lower arrival punctuality. Launceston experienced the largest decline, falling from 84.2% to 61.5%, while Melbourne declined by 21.6 percentage points.

The network improved to 70.6% in 2023, with 29 of 36 comparable airports recording higher OTP. Proserpine showed the largest recovery, increasing from 68.0% to 77.7%. However, this recovery was not universal: Hamilton Island and Mackay declined by 6.6 and 6.0 percentage points, respectively. Longer-term patterns were also visible in the heatmap. Port Lincoln achieved at least 85% arrival OTP in 12 of 14 complete years, whereas Port Macquarie remained below 80% in 13 of 14 years. The heatmap therefore distinguishes persistent airport-level patterns from network-wide temporal changes.

4.4 Monthly Performance in 2023

Figure 5 reveals substantial monthly variation in airport punctuality during 2023. Network arrival OTP declined from 76.7% in January to 69.0% in June and 68.2% in July, before falling further to 64.1% in November and 63.6% in December. December also recorded the year's highest network cancellation rate at 4.96%. The decline was geographically broad rather than restricted to isolated airports. Only

three airports recorded arrival OTP below 70% in January, compared with 18 in July, 26 in November, and 21 in December.

Several high-volume airports were also affected: Brisbane recorded 56.3% arrival OTP in December, Perth 57.0% in July, Melbourne 57.7% in November, and Sydney 60.7% in November. These patterns indicate widespread late-year weakness and airport-specific monthly irregularities. However, the 2023 monthly view alone is insufficient to establish a recurring seasonal cycle across multiple years.

4.5 Route and Airline Punctuality and Reliability

Figure 6 shows a close alignment between departure and arrival performance at route level. Using the 2023 network benchmarks of 71.0% departure OTP and 70.6% arrival OTP, 45 routes exceeded both benchmarks, while 60 routes fell below both; only 21 routes showed mixed departure and arrival performance. High flight volume did not guarantee strong reliability. Melbourne–Sydney, the highest-volume route with 24,222 sectors flown, recorded 72.5% departure OTP, 67.6% arrival OTP, and an 8.95% cancellation rate. Route direction also produced substantial differences. Darwin–Adelaide recorded a 12.5 percentage-point positive arrival-minus-departure OTP gap, whereas Adelaide–Darwin showed a 12.3-point negative gap.

Airline comparisons in Figure 7 further demonstrate variation in punctuality and reliability. QantasLink achieved the highest departure OTP at 76.4%, while Rex Airlines recorded a similar 76.3% and the lowest cancellation rate among the displayed airlines at 1.85%. In contrast, Bonza recorded 62.6% departure OTP and a 12.7% cancellation rate, although its 1,094 sectors flown indicate a substantially smaller operational scale. These findings reinforce the need to interpret punctuality

together with cancellations and flight volume rather than through OTP rankings alone.

5. Critical Evaluation and Related Analytical

Methods

5.1 Advantages and Limitations of the Selected Visualisations

The selected visualisations provide complementary analytical strengths. The proportional-symbol map preserves spatial context and jointly represents airport location, activity, and punctuality; however, overlapping bubbles can obscure smaller airports and the map is less suitable for precise ranking. Horizontal bars support more accurate airline comparison but remove geographic relationships. Line charts preserve temporal continuity and clearly reveal changes and irregularities, whereas plotting all airports as separate lines would create substantial overplotting. The heatmaps address this limitation by compressing many airport–time combinations into a comparable colour matrix, although exact values are less immediately visible without tooltips. Similarly, the route scatter plot reveals relationships, clusters, scale, and unusual observations across several measures, but combining position, size, and colour increases cognitive demand. These design choices are consistent with the overview, filtering, and details-on-demand principles of interactive information seeking described by Shneiderman (1996) and the interactive visual analysis framework discussed by Heer and Shneiderman (2012).

5.2 Scalability for Large Multidimensional Data

The methods used in this project are applicable to larger multidimensional datasets, but scalability depends on controlling both computational and perceptual complexity. The analytical layers and weighted aggregation reduce the number of records presented in each view; directly displaying all 59,472 airline–route–month observations would instead increase overplotting, visual clutter, and interaction cost. For substantially larger datasets, pre-aggregation, binning, sampling, and database-backed querying could limit the volume transferred to the visual layer. Liu et al. (2013), for example, demonstrated interactive large-data exploration using binned plots and multivariate data tiles. Progressive querying or rendering could also prioritise an approximate overview before finer detail is retrieved. Therefore, increasing record volume does not necessarily improve analytical visibility; scalable visual analytics requires data reduction and interaction strategies that preserve relevant patterns while avoiding unnecessary marks.

5.3 Supporting Analytical and Automated Techniques

The reproducible R workflow used in this project already supports automated schema inspection, duplicate detection, transformation, and validation before visualisation. Additional analytical methods could extend the exploratory process. Clustering could group airports or routes with similar profiles across punctuality, cancellation, and flight volume, while principal component analysis could reduce a larger set of correlated numerical measures into fewer dimensions for visual exploration (Pearson, 1901). Anomaly-detection methods could systematically flag unusual airline–route–month observations; for example, Isolation Forest explicitly isolates anomalous observations rather than modelling normal instances (Liu et al.,

2008). Time-series methods could further assess persistent trends and recurring seasonal structure across multiple years. These automated techniques should complement rather than replace interactive visual analysis, because statistical flags still require contextual interpretation. The present dataset also lacks weather, congestion, staffing, aircraft availability, delay duration, and disruption-cause variables. Consequently, observed irregularities remain descriptive rather than causal, while the partial 2024 coverage and monthly aggregation restrict complete annual and flight-level interpretation.

6. Conclusion

This project applied multidimensional visual analytics to examine Australian airline punctuality and reliability across network, airport, temporal, route, and airline levels. The analysis identified a strong concentration of flight activity at major airports, a broad deterioration in punctuality during 2022 followed by uneven improvement in 2023, and substantial monthly weakness toward the end of 2023. Route and airline comparisons further showed that high operational volume or punctuality alone does not necessarily indicate stronger reliability, particularly when cancellation rates and directional differences are considered. By integrating geography, time, flight volume, departure and arrival OTP, and cancellations, the analysis demonstrates the value of complementary interactive visualisations supported by reproducible weighted calculations. Future work could combine these methods with anomaly detection and additional operational variables to move from descriptive pattern identification toward more explanatory analysis.

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